

One Dozen of the Best Generic Bends

Over the nearly forty years I've been bending, a number of very powerful "generic" bends have surfaced—bends that can be applied to many different circuits and with great results. Collected together in this appendix are one dozen of the best of these bends. All are extremely simple to understand and outrageously easy to implement. If you study these bends a little, until you grasp what each is up to, they'll spring into your memory as you explore circuits, cracking your new instrument possibilities wide open.

Many of the principles in this appendix have been touched upon in the main text of this book, and it is there you'll find more information. Here, however, you'll see things packaged "to go," for a quick reference and a jump-start into action. These are the "secrets of bending" that people talk about. But they're really just commonsense applications of the anti-theory and true theory techniques that hold bending together, finally brought out into the light, in their most basic form. Very powerful stuff. Have fun!

appendix



1. Line Output

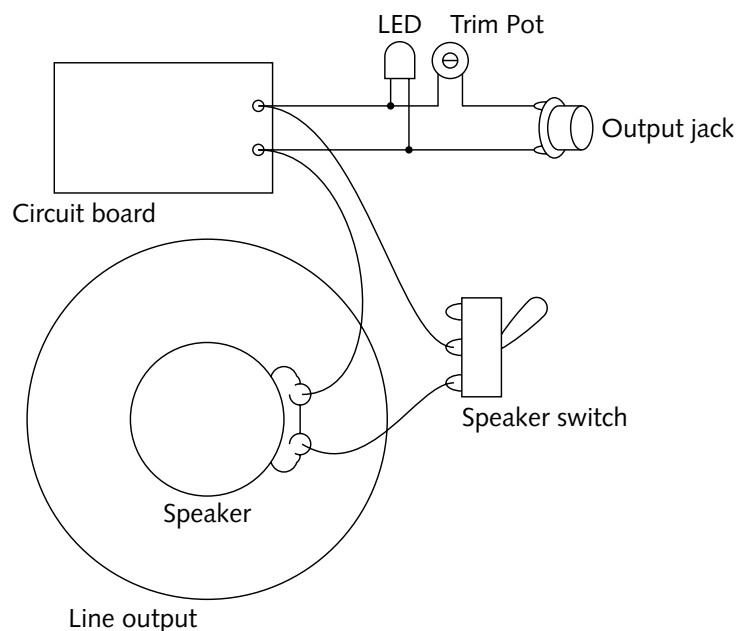


FIGURE A-1: Use this generic wiring scheme to derive a line output from your built-in speaker. The LED might not light (try various LEDs), and you might not need the trimmer. The speaker switch allows you to turn the speaker off, good for studio work. Be aware that switching the speaker off may increase the power of the line output. A nice variation on the output wiring is to use a full-sized potentiometer instead of a trim pot. Mount the full-sized pot right on the instrument and you have a handy output level control. Keep track of your polarity throughout the wiring. Use any style output jack desired.

2. "1 to X" Bend

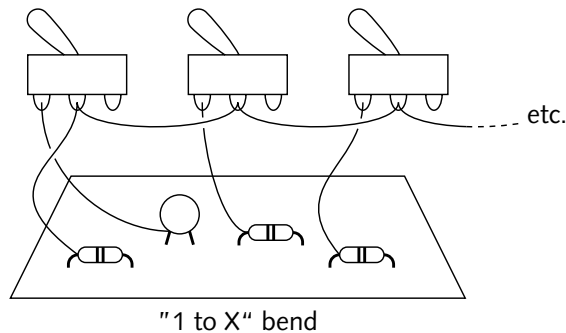


FIGURE A-2: Many of the circuits in this book look to the "1 to X" bend for their basic bent functions. The "X" number is the direct result of how many good bends the traveling end of your bending probe uncovers as it searches the circuit. Accordingly, the "1" in the "1 to X" equation represents the circuit point that the stationary end of the probe rested on while the traveling end of the probe made its tour. Any number of points may be discovered by the traveling end; the final number of switches needed to implement the "X" side of things often exceeds the area available to mount them. Also see the patch bay drawing in this appendix.

3. “X to X” Bend

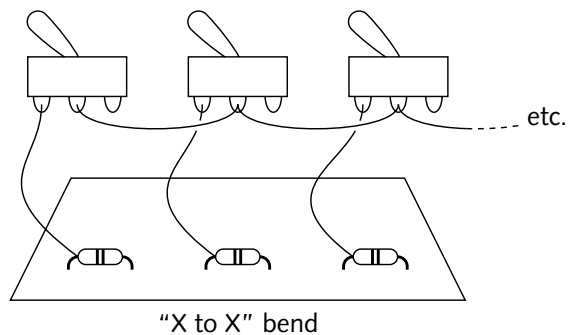
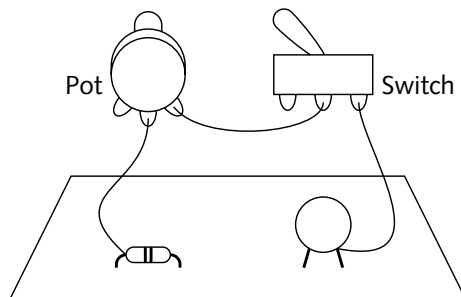


FIGURE A-3: As seen in the Harmonic Window project, this variant of the “1 to X” bend is useful for connecting many points together, but this time without there being a set single point common to all switches (point “1” in the “1 to X” bend). Within the “X to X” bend, the “1” from the “1 to X” bend simply becomes a variable. If all switches are turned off, no connection is made to the circuit. But if any of the switches are turned on, that switch then becomes the “1” in the traditional “1 to X” bend: any further turning on of switches will connect their circuit points to the point of the switch (the variable “1” switch) turned on prior. This is a cool elaboration on the traditional “1 to X” bend, often allowing a wider range of bent responses.

4. Bending Through a Potentiometer



Bending through a Potentiometer

FIGURE A-4: Any time you discover a nice bend with your bending probe, remember that that bend might be variable if you send that connection through a potentiometer. There is no set potentiometer value to try here. Large value pots (above 1M) may be too coarse in their resistance change for a bend that needs only a slight resistance change to shine. On the other hand, a small value pot (5K or less) might not present enough change in resistance to alter the bend to the degree needed. I usually begin with a 1M potentiometer. If the bent response cuts off near the bottom (low resistance) of the pot's range, I know I need a pot with a smaller resistance value (like a 10K). If I turn the pot to full resistance and the bent effect is still changing nicely, I try a pot with a larger value (like a 10M). The switch in the diagram allows the potentiometer bend to be turned on or off, as usual.

5. Two-Way Choice

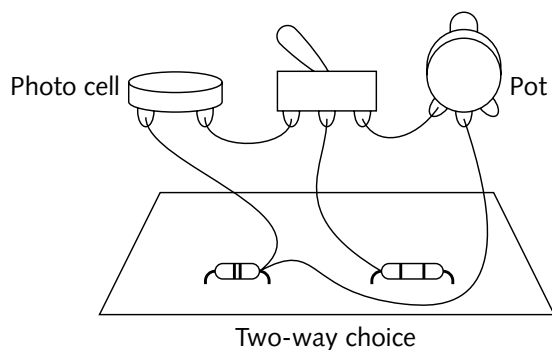


FIGURE A-5: You might remember that I discussed the exchangeability of toggle switches back in the introduction to the projects chapters. So, what *do* you do with a three-pole toggle switch if you want to use all three poles instead of using only two to duplicate the action of a two-pole (SPST) switch? As seen in the diagram, a three-pole toggle switch (SPDT) is designed to allow a signal to be directed down one of two available paths. The signal enters the center terminal of the three and exits through one of the two remaining, depending upon which way the handle of the switch is thrown. This allows you to select one of two components to run your bend through. In the diagram you'll see that the bend can now be switched between two variable resistors: a potentiometer or a photo cell. Any two components can be used.

6. Pitch Dial and Body-Contacts

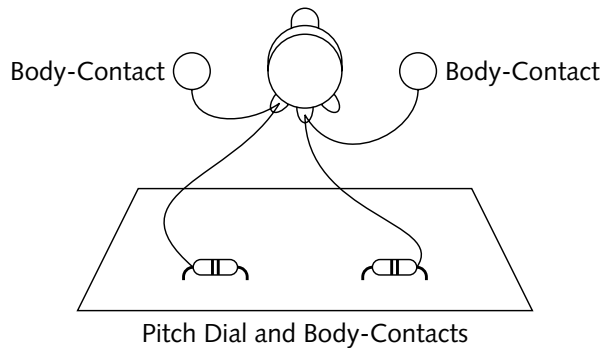


FIGURE A-6: A very common way to set the “speed” of a circuit is by means of a well-placed resistor. The resistor governs the flow of electricity from one point to another, thereby establishing overall frequency response, or pitch. But are there other points on the board that will affect overall circuit speed, or pitch, if connected to each other? As you’ve discovered via body-contacting the circuit, yes, there are. As you’ve learned, wiring a potentiometer between such body-contact points often provides such a pitch control. But does this mean the body-contact response is overthrown by the introduction of a potentiometer? Not at all. As shown in the diagram, wiring body-contacts to the lugs of the potentiometer allows you the advantage of both. You can set overall pitch via the pot, and you can still touch the body-contacts to induce real-time vibrato and pitch-bending.

7. Chance Trigger

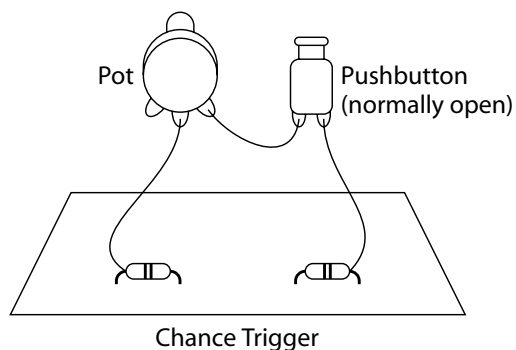


FIGURE A-7: Not all bends need to be kept connected to produce an interesting bent response. As you explore your circuits you'll eventually find two points that when briefly connected will initiate an on-going disruption of circuit activity (as seen in the SA-2 Aleatron in the projects chapters). The brief connection needed to create this response must be of the correct power to best initiate the unusual audio activity. Too much juice flowing through the connection equals a crash; too little juice, and nothing happens at all. Notice the potentiometer in the drawing. This pot regulates the amount of current flowing through the bend. Once it has been set correctly, a quick tap of the partner push button switch (a normally open mini switch) will apply the needed mini-zap to set the bent circuit response off.

8. Trimmed Potentiometer

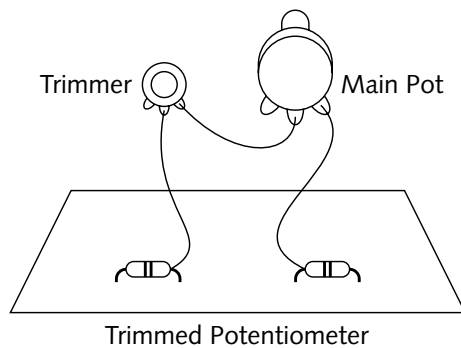


FIGURE A-8: Don't be surprised if you find a cool spot for a potentiometer to either adjust pitch or to vary some other interesting effect, only to discover that when turned all the way down (presenting no resistance to the circuit) the circuit quickly crashes. What's needed is to ensure that there's still a little resistance left in the bend when the pot's turned all the way down—to zero resistance. As shown in the diagram, the answer is simple: Once the trimmer is installed in line with the main potentiometer, all that's left is to set the trimmer to the resistance needed to keep the circuit from "seeing" zero resistance and crashing when the pot is turned all the way down. Turn the trimmer up all the way; turn the main pot down. With the bend running, turn the trimmer down until just pre-crash. This sets a safe operational minimum resistance for the bend.

9. Audition/Hold Array

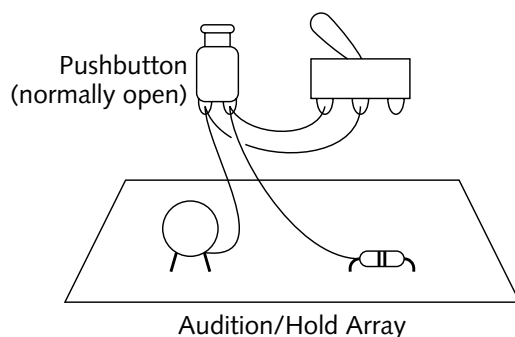


FIGURE A-9: If you've built an Incantor, you're familiar with how the looping controls work. You use the pushbutton to audition loops, pressing it over and over again to hear the loops available. Once you find a loop you want to work with, you throw the companion toggle switch to lock the loop in place. But beyond this, in simply playing notes or sequences of bent sounds, it's often nice to be able to choose between momentary or continuous operation of such a sound (as seen in the curved bottom row of switches on the Harmonic Window instrument shown on the cover of this book). Wired exactly the same as the looping matrix in an Incantor, this paired pushbutton/toggle switch combination increases the flexibility of sound production and performance.

10. Resistor Replacement

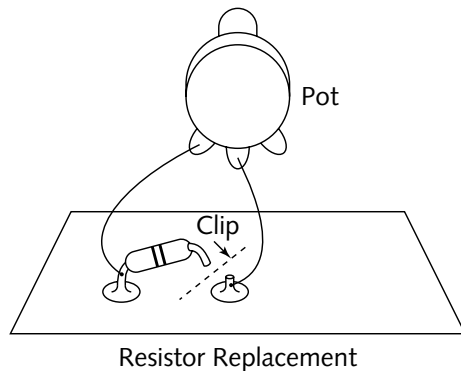
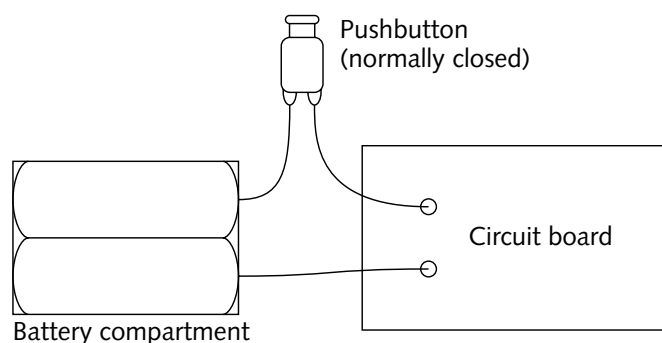


FIGURE A-10: Resistors within a circuit, while being a very basic electronic component, are responsible for keeping much complex circuit activity within balance. Replacing a resistor with one of a different value can shift circuit activity from norm to alien in a moment. So imagine what replacing a resistor with a potentiometer or photo cell can do! Replacing a resistor need not be a hassle. In fact, complete physical removal is usually unnecessary to electronically remove the resistor from the circuit. In the diagram you see a resistor with one of its leads snipped just above the board. Even though still attached to the board, the resistor is now “invisible” to the circuit because electricity can no longer travel through it. Soldering your replacement component between the still-intact lead and the circuit board stub of the original resistor’s clipped lead effectively bypasses the resistor, and with a minimum of hassle. If the substitution doesn’t work out, re-soldering the single clipped lead of the target resistor will bring the circuit back to normal for further bending.

11. Reset Button



Reset Button

FIGURE A-11: Many bent circuits crash under the new electronic pressures of circuit-bending. Simply turning the main power switch off and back on might remedy such a crash. However, there are times when interrupting the current flowing from the actual power supply is the only way to reset a deeply crashed circuit. The solution is simple if the battery compartment is connected to the circuit by means of two wires, as is usually the case. Instead of removing a battery and again replacing it in order to interrupt the power supply and therefore reset a deep crash, simply breaking the circuit between the battery compartment and the circuit board is the answer. In the illustration you'll see depicted a normally closed pushbutton switch in the middle of one of the battery compartment wires. Such a switch on either of the two wires will interrupt the power supply current when pressed, instantly resetting the crash.

12. Patch Bay

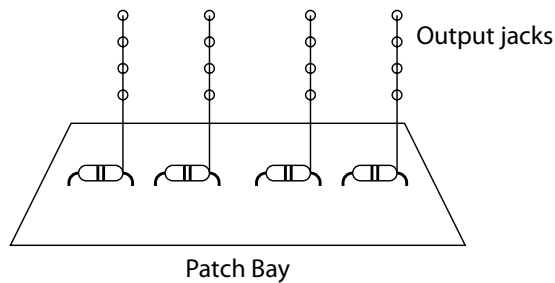


FIGURE A-12: Instead of the X to 1 or X to X bends seen in this appendix and implemented in various of the projects in this book, a simple patch bay will also allow you the same, and even greater, connection capabilities. In the drawing only four circuit points are brought forward into patch points, and these points are each expanded to only four outputs within the patch bay. Naturally, any number of circuit points can be extended into a patch bay, and each point can open up into as many outputs as desired. If you're not pressured for space you can go with the traditional "banana" jacks and plugs for designing your bay. If space is limited (as it often is within circuit-bending) the alternative is the "pin" jack, a smaller and a little less sturdy substitute. In either example, parts are available for easily making your own patch cords and bay, one of the simplest ways to implement multi-connectivity within bending.